

# Calibration Angles

| Deg    | Rad  | X_Cos     | Y_Sin     |
|--------|------|-----------|-----------|
| 0.00   | 0.00 | 1.000000  | 0.000000  |
| 35.00  | 0.61 | 0.819152  | 0.573576  |
| 60.00  | 1.05 | 0.500000  | 0.866025  |
| 90.00  | 1.57 | 0.000000  | 1.000000  |
| 120.00 | 2.09 | -0.500000 | 0.866025  |
| 145.00 | 2.53 | -0.819152 | 0.573576  |
| 180.00 | 3.14 | -1.000000 | 0.000000  |
| 210.00 | 3.67 | -0.866025 | -0.500000 |
| 240.00 | 4.19 | -0.500000 | -0.866025 |
| 270.00 | 4.71 | 0.000000  | -1.000000 |
| 300.00 | 5.24 | 0.500000  | -0.866025 |
| 330.00 | 5.76 | 0.866025  | -0.500000 |

55.00  
117.50

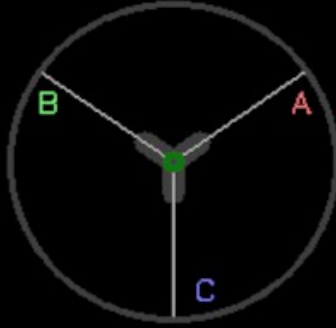
## PWM to Percentage

| PWM     | Percent |         |     |
|---------|---------|---------|-----|
| 2047.00 | 0.00    | 2047.00 | 0.0 |
| 1920.00 | 0.94    | 1920.00 | 0.1 |
| 1024.00 | 0.50    | 1024.00 | 0.5 |
|         |         | 500.00  | 0.8 |
|         |         | 0.00    | 1.0 |
| 0.00    | 100.00  |         |     |

| Marker Visibility |      |      |      |      |      |      |      |      | Telemetry |           |      | Task State |     |     |     |     | Serial Communications  |          |                |     |       |      |     |      |     |      |     |       |      |     |        |     |      |     |      |      |      |     |
|-------------------|------|------|------|------|------|------|------|------|-----------|-----------|------|------------|-----|-----|-----|-----|------------------------|----------|----------------|-----|-------|------|-----|------|-----|------|-----|-------|------|-----|--------|-----|------|-----|------|------|------|-----|
| 1.00              | 2.00 | 3.00 | 4.00 | 5.00 | 6.00 | 7.00 | 8.00 | 9.00 | Active:   |           | [ID] |            |     |     |     |     | Connected?             |          | Teensy State   |     | In    | PwmA |     | PwmB |     | PwmC |     | Out   | EncA |     | EncA   |     | EncA |     | CurA | CurB | CurC |     |
| ###               | ###  | ###  | ###  | ###  | ###  | ###  | ###  | ###  | x         | [POS_X]   |      |            |     |     |     |     | [Yes/No]               |          | [STATE]        |     | [ctr] |      |     |      |     |      |     | [ctr] |      |     |        |     |      |     |      |      |      |     |
| ###               | ###  | ###  | ###  | ###  | ###  | ###  | ###  | ###  | y         | [POS_Y]   |      |            |     |     |     |     |                        |          |                |     |       |      |     |      |     |      |     |       |      |     |        |     |      |     |      |      |      |     |
| ###               | ###  | ###  | ###  | ###  | ###  | ###  | ###  | ###  | z         | [POS_Z]   |      |            |     |     |     |     |                        |          |                |     |       |      |     |      |     |      |     |       |      |     |        |     |      |     |      |      |      |     |
| ###               | ###  | ###  | ###  | ###  | ###  | ###  | ###  | ###  | R2        | [POS_XY]  |      |            |     |     |     |     |                        |          |                |     |       |      |     |      |     |      |     |       |      |     |        |     |      |     |      |      |      |     |
| tive Tag          |      |      |      |      |      |      |      |      | R3        | [POS_XYZ] |      |            |     |     |     |     | Keyboard Input Monitor |          |                |     |       |      |     |      |     |      |     |       |      |     |        |     |      |     |      |      |      |     |
|                   |      |      |      |      |      |      |      |      | Roll      | [ROLL]    |      |            |     |     |     |     | Buffer                 | [BUFFER[ | [DISPLAY TEXT] |     |       |      |     |      |     |      |     |       | Last |     | [LAST] |     |      |     |      |      |      |     |
| 1.00              | 2.00 | 3.00 | 4.00 | 5.00 | 6.00 | 7.00 | 8.00 | 9.00 | ###       | ###       | ###  | ###        | ### | ### | ### | ### | ###                    | ###      | ###            | ### | ###   | ###  | ### | ###  | ### | ###  | ### | ###   | ###  | ### | ###    | ### | ###  | ### | ###  | ###  | ###  | ### |

| <div></div>    |    |    |    |               |    |    |    |    |    |    |                 | <div></div> |     |     |     |                       |            |     |      |     |       |      |      |      |      |      |      |      |  |  |  |  |  |  |  |
|----------------|----|----|----|---------------|----|----|----|----|----|----|-----------------|-------------|-----|-----|-----|-----------------------|------------|-----|------|-----|-------|------|------|------|------|------|------|------|--|--|--|--|--|--|--|
| Target Markers |    |    |    | World Markers |    |    |    |    |    |    | Active Marker 1 |             |     |     |     | Serial Communications |            |     |      |     |       |      |      |      |      |      |      |      |  |  |  |  |  |  |  |
| 01             | 02 | 03 | 04 | —             | 82 | 85 | —  | 86 | 83 | —  | Position        |             |     |     |     |                       | Port       |     | Type | IDX | State | PwmA | PwmB | PwmC | EncA | EncB | EncC | Rev? |  |  |  |  |  |  |  |
| 05             | 06 | 07 | 08 | 91            | 89 | 80 | 84 | 81 | 90 | 92 | X               | Y           | Z   | R2  | R3  |                       | Serial Out | Off | 0    | 0   | 0     | 0    | 0    | 0    | —    | —    | —    | 0    |  |  |  |  |  |  |  |
| 09             | 10 | 11 | 12 | —             | 78 | 87 | —  | 88 | 79 | —  | 81              | 80          | 311 | 114 | 331 |                       | Serial In  | Off | 0    | 0   | 0     | 0    | 0    | 0    | 0    | 0    | 0    | 0    |  |  |  |  |  |  |  |
| Status:        |    |    |    |               |    |    |    |    |    |    |                 |             |     |     |     |                       |            |     |      |     |       |      |      |      |      |      |      |      |  |  |  |  |  |  |  |

| System Status |     |
|---------------|-----|
| Clock         |     |
| Camera        |     |
| State         |     |
| Teensy        |     |
| UserID        | --- |
| Mouse         | x   |
| State         | y   |

| TELEMETRY |               |          |         | CONTROLLER |     |             |      |           |      |            |       | AMPLIFIER     |          |         |       |      | TASK MONITOR |                                                                                       |      |                 |         |     |   |   |   |
|-----------|---------------|----------|---------|------------|-----|-------------|------|-----------|------|------------|-------|---------------|----------|---------|-------|------|--------------|---------------------------------------------------------------------------------------|------|-----------------|---------|-----|---|---|---|
| 1         | Position      | Velocity | Integr. | Freq       | Kp  | Ki          | Kd   |           | Kp+E | Ki+IE      | Kp+dE | Total         | Released |         | A     | B    | C            |  | Name | No task running | User ID | 123 |   |   |   |
|           | [mm]          | [mm/s]   | [mm]    | 52         | [u] | [u]         | [u]  |           | [u]  | [u]        | [u]   | [u]           | Tension  | [%]     | 0.0   | 0.0  | 0.0          |                                                                                       | Time | 0.00            |         |     |   |   |   |
| x         | 0             | 0        | 0       | ABD        | 0.0 | 0.0         | 0.00 | x         | 0.0  | 0.0        | 0.0   | 0.0           | Drive    | [%]     | 0.0   | 0.0  | 0.0          |                                                                                       |      |                 |         |     |   |   |   |
| y         | 0             | 0        | 0       | ADD        | 0.0 | 0.0         | 0.00 |           | 0.0  | 0.0        | 0.0   | 0.0           | Total    | [%]     | 0.0   | 0.0  | 0.0          |                                                                                       |      |                 |         |     |   |   |   |
| z         | 0             | 0        | 0       | FLEX       | 0.0 | 0.0         | 0.00 | y         | 0.0  | 0.0        | 0.0   | 0.0           | Max      | [%]     | 20.0  | 20.0 | 20.0         |                                                                                       |      |                 |         |     |   |   |   |
| Rxy       | 0             | 0        | xx      | EXT        | 0.0 | 0.0         | 0.00 |           | 0.0  | 0.0        | 0.0   | 0.0           | PWM      | [dc]    | 2044  | 2044 | 2044         |                                                                                       |      |                 |         |     |   |   |   |
| Rxyz      | 0             | 0        | xx      | No Limit   |     | Logging Off |      | Teensy In |      | Teensy Out |       | Amplifier Off |          | Current | [A]   | 0.00 | 0.00         |                                                                                       | 0.00 | Touchscreen     |         | x   | 0 | y | 0 |
| PC Out    | Not sending   |          |         |            |     |             |      |           |      |            |       | Lag           |          | Angle   | [deg] | 0.00 | 0.00         |                                                                                       | 0.00 |                 |         |     |   |   |   |
| PC In     | Not receiving |          |         |            |     |             |      |           |      |            |       | 0             | [ms]     | Limit   | [deg] | 0.00 | 0.00         | 0.00                                                                                  |      |                 |         |     |   |   |   |

| System State            |          |         |   |                 |       |         |       |  |  |
|-------------------------|----------|---------|---|-----------------|-------|---------|-------|--|--|
| PC State                | [STATE]  |         |   | Tension         |       |         | [SET] |  |  |
| Camera                  | [FREQ]   |         |   | Home Position   |       |         | [SET] |  |  |
| Serial                  | [FREQ]   |         |   | Cal1: ARoM      |       |         | [SET] |  |  |
| Teensy                  | [STATE]  |         |   | Cal2: Stiffness |       |         | [SET] |  |  |
| Amplifiers              | [STATE]  |         |   | Cal3: Offset    |       |         | [SET] |  |  |
| Target Telemetry        |          |         |   |                 |       |         |       |  |  |
|                         | x        | y       | z | Rxy             | Rxyz  |         |       |  |  |
| Position [mm]           |          |         |   |                 |       |         |       |  |  |
| Velocity [mm/s]         |          |         |   |                 |       |         |       |  |  |
| Accumulated [mm]        |          |         |   |                 |       |         |       |  |  |
| Controller Output       |          |         |   |                 |       |         |       |  |  |
|                         |          | Motor A |   | Motor B         |       | Motor C |       |  |  |
| Tension                 | [N]      |         |   |                 |       |         |       |  |  |
|                         | [PWM]    |         |   |                 |       |         |       |  |  |
| Force                   | [N]      |         |   |                 |       |         |       |  |  |
|                         | [PWM]    |         |   |                 |       |         |       |  |  |
| Stiffness               | [N/mm]   |         |   |                 |       |         |       |  |  |
| KP                      | [N/mm]   |         |   |                 |       |         |       |  |  |
| KI                      | [N/mm*s] |         |   |                 |       |         |       |  |  |
| Commanded               | [N]      |         |   |                 |       |         |       |  |  |
|                         | [I]      |         |   |                 |       |         |       |  |  |
|                         | [PWM]    |         |   |                 |       |         |       |  |  |
| Virtual Fingertip       |          |         |   |                 |       |         |       |  |  |
| Tendons                 |          | Motor A |   | Motor B         |       | Motor C |       |  |  |
| Unspooled Angle         |          |         |   |                 |       |         |       |  |  |
| Home Angle              |          |         |   |                 |       |         |       |  |  |
| Effective Pulley Radius |          |         |   |                 |       |         |       |  |  |
| Tendon Length Change    |          |         |   |                 |       |         |       |  |  |
| <div>#### ##</div>      |          |         |   |                 |       |         |       |  |  |
| Virtual Fingertip       |          | [xxx]   |   |                 | [yyy] |         |       |  |  |

|              |           |
|--------------|-----------|
| PC State     | [STATE]   |
| Camera       | [FREQ]    |
| Serial       | [FREQ]    |
| Teensy State | [T_STATE] |
| Amplifiers   | [STATE]   |
| Tension      | [STATE]   |
| ARoM         | [STATE]   |
| Stiffness    | [STATE]   |
| Offset       | [STATE]   |

#VALUE!

|              |         |
|--------------|---------|
| Center       | 225.00  |
| Radius       | 200.00  |
|              |         |
| Panel Height | 1344.00 |
| Rows         | 42.00   |
| Row Height   | 32.00   |

System Readiness

|                 |       |               |         |
|-----------------|-------|---------------|---------|
| PC State        | IDLE  | Tension       | Not Set |
| Camera          | 88 Hz | Home Position | Not Set |
| Serial          | 0 Hz  | C1: ARoM      | Not Set |
| Teensy State    | --    | C2: Stiffness | Not Set |
| Amplifier State | --    | C3: Offset    | Not Set |

Target Telemetry / Error Vector

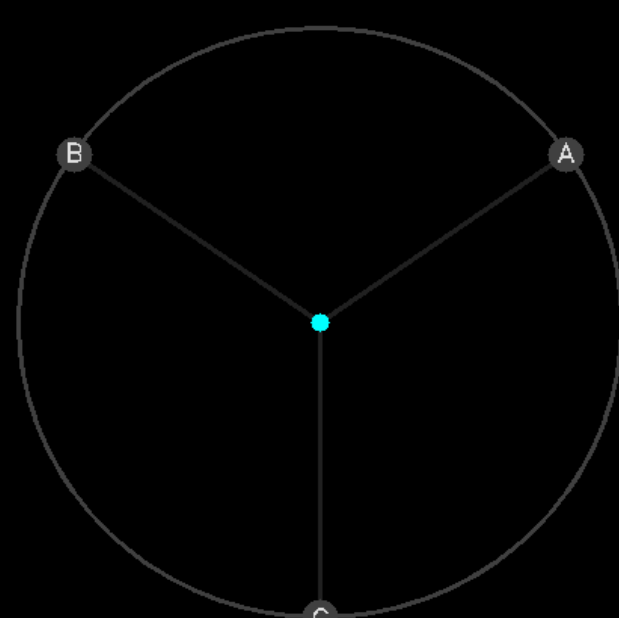
|                  | x   | y   | z  | Rxy | Rxyz |
|------------------|-----|-----|----|-----|------|
| Position [mm]    | --  | --  | -- | --  | --   |
| Velocity [mm/s]  | 0.0 | 0.0 | -- | 0.0 | --   |
| Accumulated [mm] | 0.0 | 0.0 | -- | 0.0 | --   |

Controller Output

|           |        | A   AdEx | B   AbEx | C   Flex |
|-----------|--------|----------|----------|----------|
| Tension   | N      | 0.10     | 0.10     | 0.10     |
|           | PWM    | 2028     | 2028     | 2028     |
| Force     | N      | 0.00     | 0.00     | 0.00     |
|           | PWM    | 2047     | 2047     | 2047     |
| Stiffness | N/mm   | --       | --       | --       |
| Gain kP   | N/mm   | 0.00     | 0.00     | 0.00     |
| Gain kI   | N/mm*s | --       | --       | --       |
| Output    | N      | 0.10     | 0.10     | 0.10     |
|           | I      | --       | --       | --       |
|           | PWM    | 2028     | 2028     | 2028     |

Virtual Fingertip Mapping

| Tendons                 | A   AdEx | B   AbEx | C   Flex |
|-------------------------|----------|----------|----------|
| Unspooled Angle         | 0.00     | 0.00     | 0.00     |
| Home Angle              | 0.00     | 0.00     | 0.00     |
| Effective Pulley Radius | 2.5      | 2.5      | 2.5      |
| Length Change           | 0.0      | 0.0      | 0.0      |



Virtual Position
0.0
0.0

TELEMETRY

| 1    | Position [mm] | Velocity [mm/s] | Integr [mm] |
|------|---------------|-----------------|-------------|
| x    | 0             | 0               | 0           |
| y    | 0             | 0               | 0           |
| z    | 0             | 0               | 0           |
| Rxy  | 0             | 0               | xx          |
| Rxyz | 0             | 0               | xx          |

PC Out
Not sending

PC In
Not receiving

| TELEMETRY |               |          |        |
|-----------|---------------|----------|--------|
| 1         | Position      | Velocity | Integr |
|           | [mm]          | [mm/s]   | [mm]   |
| x         | 0             | 0        | 0      |
| y         | 0             | 0        | 0      |
| z         | 0             | 0        | 0      |
| Rxy       | 0             | 0        | xx     |
| Rxyz      | 0             | 0        | xx     |
| PC Out    | Not sending   |          |        |
| PC In     | Not receiving |          |        |

|         |      |
|---------|------|
| 1380.00 | 30.0 |
| 46.00   |      |

| Controller                      |        |               |          |          |      | — | × |
|---------------------------------|--------|---------------|----------|----------|------|---|---|
| System Readiness                |        |               |          |          |      |   |   |
| PC State                        | IDLE   | Tension       |          | Not Set  |      |   |   |
| Camera                          | 88 Hz  | Home Position |          | Not Set  |      |   |   |
| Serial                          | 0 Hz   | C1: ARoM      |          | Not Set  |      |   |   |
| Teensy State                    | --     | C2: Stiffness |          | Not Set  |      |   |   |
| Amplifier State                 | --     | C3: Offset    |          | Not Set  |      |   |   |
| Target Telemetry / Error Vector |        |               |          |          |      |   |   |
|                                 | x      | y             | z        | Rxy      | Rxyz |   |   |
| Position [mm]                   | --     | --            | --       | --       | --   |   |   |
| Velocity [mm/s]                 | 0.0    | 0.0           | --       | 0.0      | --   |   |   |
| Accumulated [mm]                | 0.0    | 0.0           | --       | 0.0      | --   |   |   |
| Controller Output               |        |               |          |          |      |   |   |
|                                 |        | A   AdEx      | B   AbEx | C   Flex |      |   |   |
| Tension                         | N      | 0.10          | 0.10     | 0.10     |      |   |   |
|                                 | PWM    | 2028          | 2028     | 2028     |      |   |   |
| Force                           | N      | 0.00          | 0.00     | 0.00     |      |   |   |
|                                 | PWM    | 2047          | 2047     | 2047     |      |   |   |
| Stiffness                       | N/mm   | --            | --       | --       |      |   |   |
| Gain kP                         | N/mm   | 0.00          | 0.00     | 0.00     |      |   |   |
| Gain kI                         | N/mm*s | --            | --       | --       |      |   |   |
| Output                          | N      | 0.10          | 0.10     | 0.10     |      |   |   |
|                                 | I      | --            | --       | --       |      |   |   |
|                                 | PWM    | 2028          | 2028     | 2028     |      |   |   |
| Virtual Fingertip Mapping       |        |               |          |          |      |   |   |
| Tendons                         |        | A   AdEx      | B   AbEx | C   Flex |      |   |   |
| Unspooled Angle                 |        | 0.00          | 0.00     | 0.00     |      |   |   |
| Home Angle                      |        | 0.00          | 0.00     | 0.00     |      |   |   |
| Effective Pulley Radius         |        | 2.5           | 2.5      | 2.5      |      |   |   |
| Length Change                   |        | 0.0           | 0.0      | 0.0      |      |   |   |
|                                 |        |               |          |          |      |   |   |
| Virtual Position                |        | 0.0           |          | 0.0      |      |   |   |

[illegible][illegible]

**ptions**

IDLE, CAL, CAL1, CAL2, CAL3,

IDLE, CAL, CAL1, CAL2, CAL3, TENSIO

ON, --  
[HZ]

ON, --

SET, OFF, --

SET, --

SET, --

SET, --

|                       |  |  |  |  |  |
|-----------------------|--|--|--|--|--|
| Active ROM            |  |  |  |  |  |
| Stiffness Calibration |  |  |  |  |  |
| Offset                |  |  |  |  |  |

| B | C |
|---|---|
|   |   |
|   |   |
|   |   |
|   |   |
|   |   |
|   |   |
|   |   |
|   |   |

**AMPLIFIER**

| C    |
|------|
| 0.0  |
| 0.0  |
| 0.0  |
| 20.0 |
| 2044 |
| 0.00 |
| 0.00 |
| 0.00 |

**TASK MONITOR**

| Name        | No task running | User ID | 123 |
|-------------|-----------------|---------|-----|
| Time        | 0.00            |         |     |
|             |                 |         |     |
| Touchscreen | x               | o       | y o |
|             |                 |         |     |

[illegible]

| Set         | Key   | DEC    |   |       |
|-------------|-------|--------|---|-------|
| Lower Alpha | a     | 97.00  | A | 65.00 |
|             | b     | 98.00  | B | 66.00 |
|             | c     | 99.00  | C | 67.00 |
|             | d     | 100.00 | D | 68.00 |
|             | e     | 101.00 | E | 69.00 |
|             | f     | 102.00 | F | 70.00 |
|             | g     | 103.00 | G | 71.00 |
|             | h     | 104.00 | H | 72.00 |
|             | i     | 105.00 | I | 73.00 |
|             | j     | 106.00 | J | 74.00 |
|             | k     | 107.00 | K | 75.00 |
|             | l     | 108.00 | L | 76.00 |
|             | m     | 109.00 | M | 77.00 |
|             | n     | 110.00 | N | 78.00 |
|             | o     | 111.00 | O | 79.00 |
|             | p     | 112.00 | P | 80.00 |
|             | q     | 113.00 | Q | 81.00 |
|             | r     | 114.00 | R | 82.00 |
|             | s     | 115.00 | S | 83.00 |
|             | t     | 116.00 | T | 84.00 |
|             | u     | 117.00 | U | 85.00 |
|             | v     | 118.00 | V | 86.00 |
|             | w     | 119.00 | W | 87.00 |
|             | x     | 120.00 | X | 88.00 |
|             | y     | 121.00 | Y | 89.00 |
|             | z     | 122.00 | Z | 90.00 |
| Symbols     | !     | 33.00  |   |       |
|             | "     | 34.00  |   |       |
|             | #     | 35.00  |   |       |
|             | \$    | 36.00  |   |       |
|             | %     | 37.00  |   |       |
|             | &     | 38.00  |   |       |
|             | '     | 39.00  |   |       |
|             | (     | 40.00  |   |       |
|             | )     | 41.00  |   |       |
|             | *     | 42.00  |   |       |
|             | +     | 43.00  |   |       |
|             | ,     | 44.00  |   |       |
|             | -     | 45.00  |   | MINUS |
|             | .     | 46.00  |   |       |
|             | /     | 47.00  |   |       |
|             | :     | 58.00  |   |       |
|             | ;     | 59.00  |   |       |
|             | <     | 60.00  |   |       |
|             | =     | 61.00  |   | PLUS  |
|             | >     | 62.00  |   |       |
|             | ?     | 63.00  |   |       |
|             | @     | 64.00  |   |       |
|             | {     | 123.00 |   |       |
|             |       | 124.00 |   |       |
|             | }     | 125.00 |   |       |
|             | ~     | 126.00 |   |       |
| Numbers     | 0     | 48.00  |   |       |
|             | 1     | 49.00  |   |       |
|             | 2     | 50.00  |   |       |
|             | 3     | 51.00  |   |       |
|             | 4     | 52.00  |   |       |
|             | 5     | 53.00  |   |       |
|             | 6     | 54.00  |   |       |
|             | 7     | 55.00  |   |       |
|             | 8     | 56.00  |   |       |
|             | 9     | 57.00  |   |       |
| Keypad      | 0.00  | 176.00 |   |       |
|             | 1.00  | 177.00 |   |       |
|             | 2.00  | 178.00 |   |       |
|             | 3.00  | 179.00 |   |       |
|             | 4.00  | 180.00 |   |       |
|             | 5.00  | 181.00 |   |       |
|             | 6.00  | 182.00 |   |       |
|             | 7.00  | 183.00 |   |       |
|             | 8.00  | 184.00 |   |       |
|             | 9.00  | 185.00 |   |       |
|             | =     | 61.00  |   |       |
|             | /     | 175.00 |   |       |
|             | *     | 170.00 |   |       |
|             | -     | 173.00 |   |       |
|             | +     | 171.00 |   |       |
|             | .     | 174.00 |   |       |
|             | (     | 40.00  |   |       |
|             | )     | 41.00  |   |       |
|             | ←     | 8.00   |   |       |
|             | Clear | 27.00  |   |       |
|             | Enter | 171.00 |   |       |
| Back        | BkSp  | 8.00   |   |       |
|             | Del   | 255.00 |   |       |
|             | Space | 32.00  |   |       |

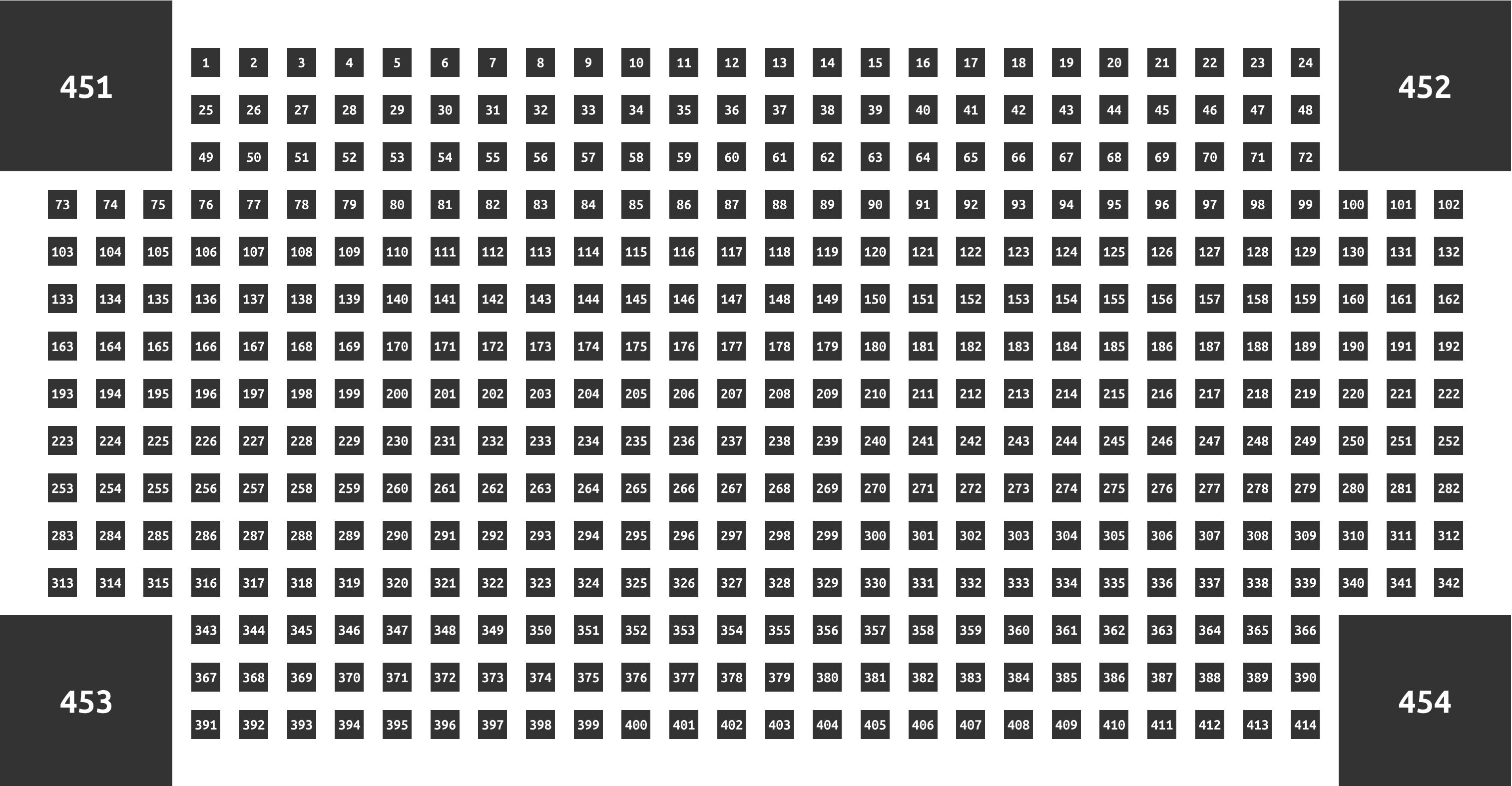


## System State

|              |           |
|--------------|-----------|
| PC State     | [STATE]   |
| Camera       | [FREQ]    |
| Serial       | [FREQ]    |
| Teensy State | [T_STATE] |
| Amplifiers   | [STATE]   |
| Tension      | [STATE]   |
| ARoM         | [STATE]   |
| Stiffness    | [STATE]   |
| Offset       | [STATE]   |

| Screen Size     |  |          |           |       |        |        |        |        |         |        |         |
|-----------------|--|----------|-----------|-------|--------|--------|--------|--------|---------|--------|---------|
|                 |  | MM       | PX        |       |        | MM2PX  | PX2MM  |        |         |        |         |
|                 |  | 527.0400 | 1920.0000 |       |        | 3.6430 | 0.2745 |        |         |        |         |
|                 |  | 296.4600 | 1080.0000 |       |        |        |        |        |         |        |         |
|                 |  |          |           |       |        |        |        |        |         |        |         |
|                 |  |          |           | Used  |        |        |        | Remain |         |        |         |
| Exclusion Zones |  |          |           | Width | Height |        |        | Width  | Height  |        |         |
|                 |  | MM       | PX        | MM    | PX     | MM     | PX     | MM     | PX      | MM     | PX      |
| Screen Edge     |  | 5.00     | 18.215    | 10.00 | 36.43  | 10.00  | 36.43  | 517.04 | 1883.57 | 286.46 | 1043.57 |
|                 |  |          |           |       |        |        |        |        |         |        |         |
| Markers         |  | MM       | PX        |       |        |        |        |        |         |        |         |
| Coarse          |  | 50.00    | 182.00    |       |        |        |        |        |         |        |         |
| Fine            |  | 8.00     | 29.00     |       |        |        |        |        |         |        |         |
|                 |  |          |           |       |        |        |        |        |         |        |         |
| Padding         |  | MM       | PX        |       |        |        |        |        |         |        |         |
| Outside         |  | 8.00     | 29.00     |       |        |        |        |        |         |        |         |





**Coarse Markers:** Size = 56.0 mm    Padding = 8.0 mm    Borders: Top = 16.50 mm, Bottom = 16.50 mm, Left = 11.5 mm, Right = 11.5 mm

**Fine Markers:** Size = 8.0 mm    Padding = 8.0 mm    Borders: Top = 32.23 mm, Bottom = 32.23 mm, Left = 27.5 mm, Right = 27.5 mm    Pitch: 16 mm center-to-center